



East West University

Department of Computer Science and Engineering

Semester : Spring 2021

Course Title: Electrical Circuits

Course Code: CSE 209

Section: 03

Experiment Number : 06

Experiment Title : Verification of Thevenin's Theorem.

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Objective : To verify the Thevenin's theorem theoretically experimentally, and using PSpice simulation.

Circuit Diagram:

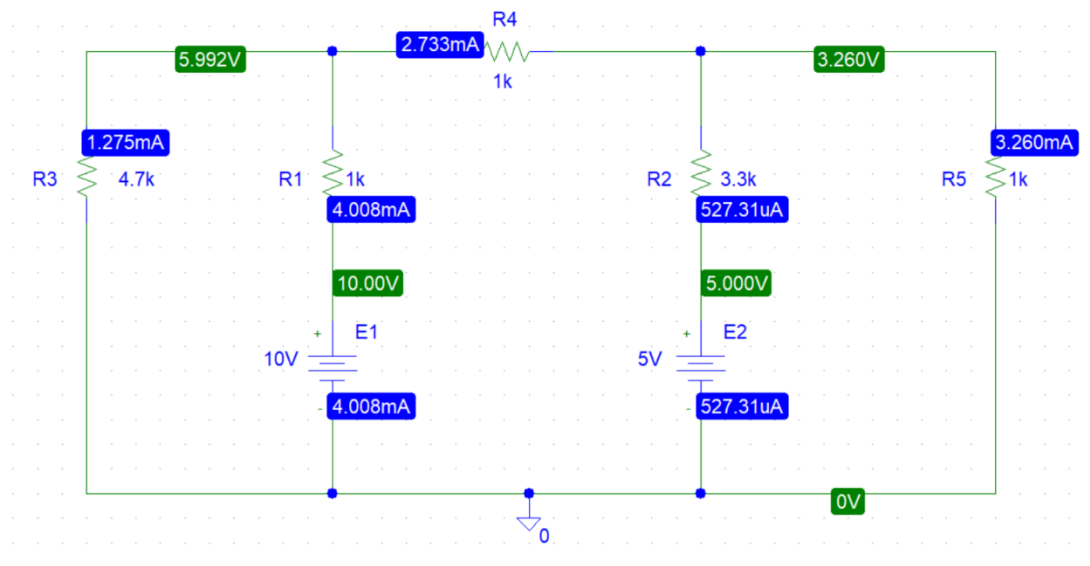


Figure -1

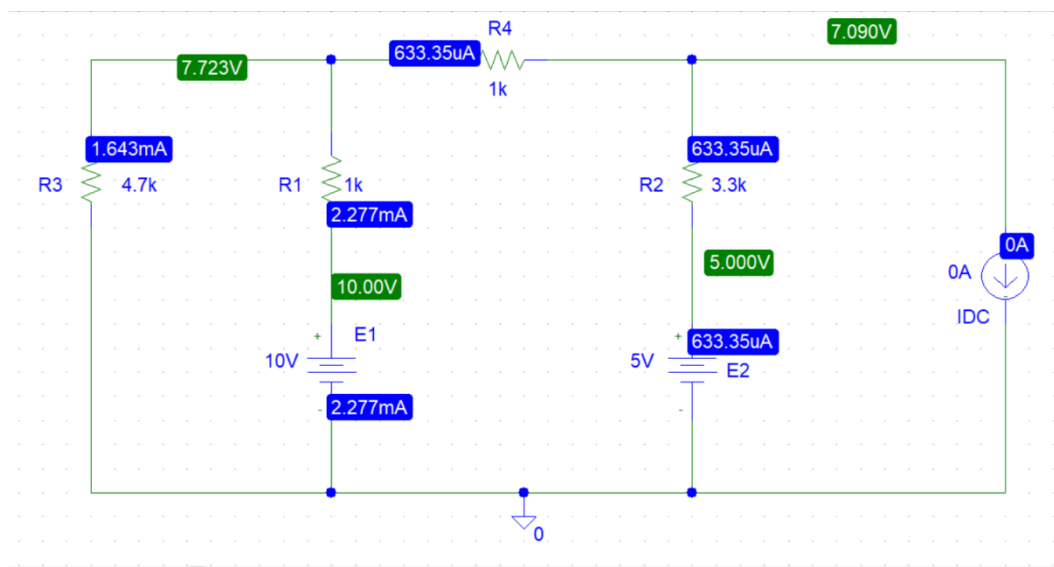


Figure-2

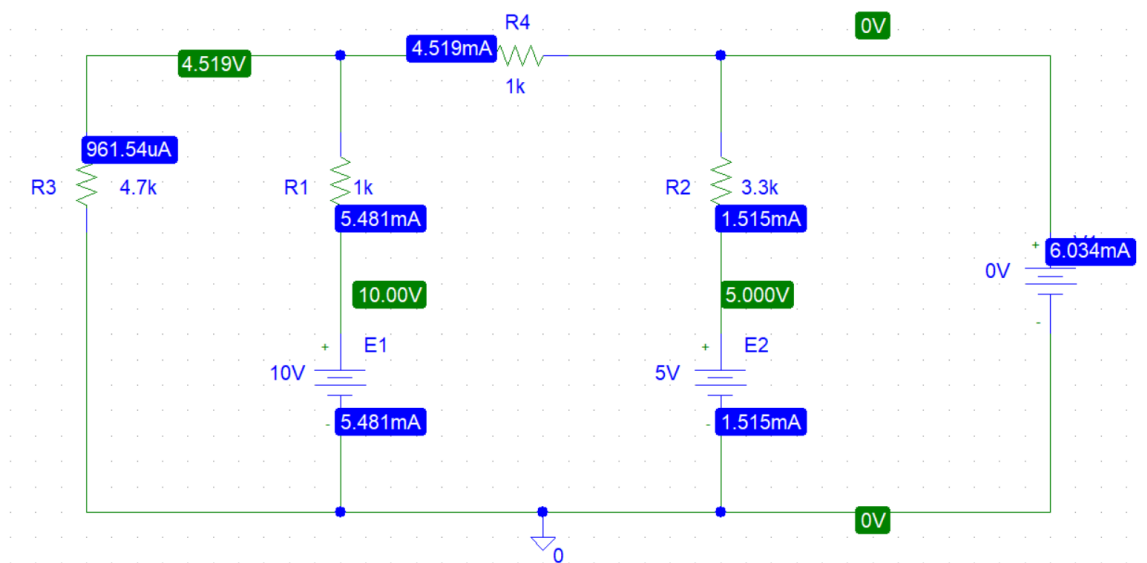


Figure-3

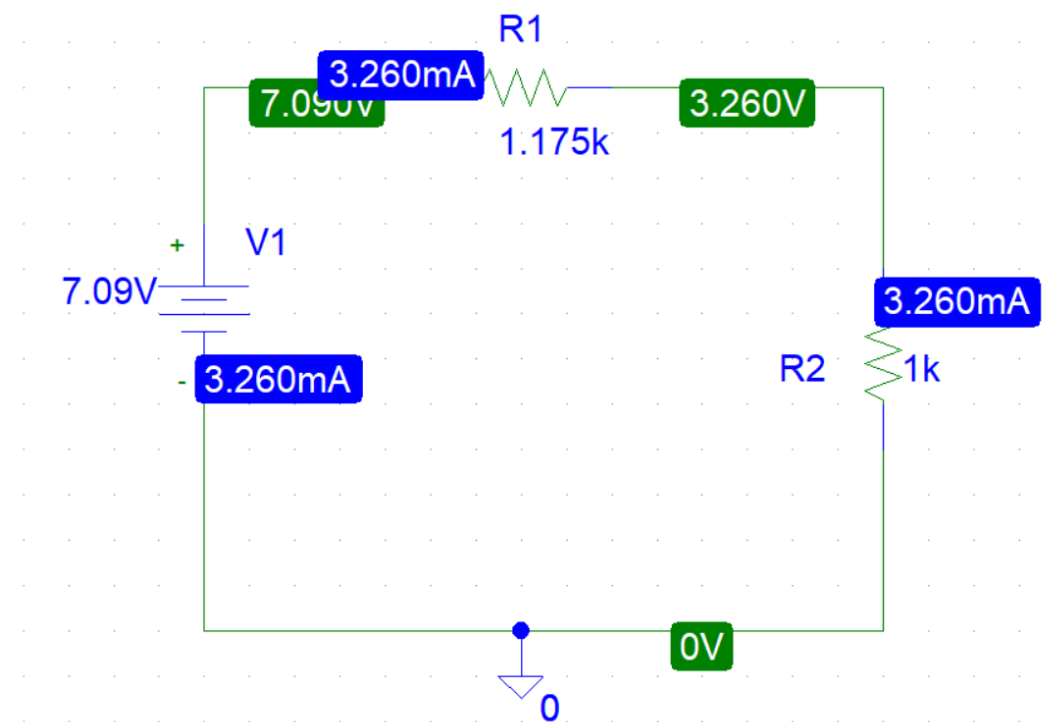


Figure -4

Experimental Datasheet:

Value of E ₁	Value of E ₂	Value of V _L	Value of I _L	Value of V _{OC}	Value of I _{SC}	Value of Resistors
10	5	3.260	3.260mA	7.090	6.03mA	R ₁ =1 kΩ R ₂ =4.7 kΩ R ₃ =3.3 kΩ R ₄ = 1 kΩ R _L = 1 kΩ

Answers to the post lab report questions:

1. For figure 1:

Applying KVL at mesh 1,

$$4.7ki_1 + 1k(i_1 - i_2) = -10$$

$$5.7ki_1 - 1k i_2 = -10 \dots\dots(1)$$

Applying KVL at mesh 2,

$$1k(i_2 - i_1) + 1ki_2 + 3.3k(i_2 - i_3) = 10 - 5$$

$$-1ki_1 + 5.3ki_2 - 3.3k i_3 = 5 \dots\dots(2)$$

Applying KVL at mesh 3,

$$3.3k(i_3 - i_2) + 1k i_3 = 5$$

$$-3.3k i_2 + 4.3k i_3 = 5 \dots\dots(3)$$

Solving equation 1,2 & 3,

$$i_1 = -1.275 \times 10^{-3} \text{A} = -1.275 \text{mA}$$

$$i_2 = 2.733 \times 10^{-3} \text{A} = 2.733 \text{mA}$$

$$i_3 = 3.260 \times 10^{-3} \text{A} = 3.260 \text{mA}$$

$$\text{here, } i_3 = I_L = 3.260 \text{mA}$$

$$\begin{aligned} V_L &= I_L \times R = 3.260 \text{mA} \times 1 \text{k}\Omega \\ &= 3.260 \text{V} \end{aligned}$$

For figure 2:

Applying KCL at node 1,

$$\frac{V_1}{4.7\text{k}} + \frac{V_1 - V_2}{1\text{k}} + \frac{V_1 - 10}{1\text{k}} = 0$$

$$2.213 \times 10^{-3} V_1 - 1 \times 10^{-3} V_2 = 0.01 \dots\dots(1)$$

Applying KCL at node 2,

$$\frac{V_2 - V_1}{1\text{k}} + \frac{V_2 - 5}{3.3\text{k}} = 0$$

$$-1 \times 10^{-3} V_1 + 1.30303 \times 10^{-3} V_2 = 1.51515 \times 10^{-3} \dots\dots(2)$$

Solving equation 1&2,

$$v_1 = 7.723 \text{V}$$

$$v_2 = 7.090 \text{V}$$

$$\text{here, } V_{OC} = V_{th} = V_2 = 7.090 \text{V}$$

For figure 3:

Applying KVL at mesh 1,

$$4.7ki_1 + 1k(i_1 - i_2) = -10$$

$$5.7ki_1 - 1k i_2 = -10 \dots\dots (1)$$

Applying KVL at mesh 2,

$$1k(i_2 - i_1) + 1ki_2 + 3.3k(i_2 - i_3) = 10 - 5$$

$$-1ki_1 + 5.3ki_2 - 3.3k i_3 = 5 \dots\dots (2)$$

Applying KVL at mesh 3,

$$3.3k(i_3 - i_2) = 5$$

$$0i_1 - 3.3k i_2 + 3.3k i_3 = 5 \dots\dots (3)$$

Solving equation 1, 2 & 3,

$$i_1 = -9.6154 \times 10^{-3} \text{A} = -0.96154 \text{mA}$$

$$i_2 = 4.519 \times 10^{-3} \text{A} = 4.519 \text{mA}$$

$$i_3 = 6.034 \times 10^{-3} \text{A} = 6.034 \text{mA}$$

here, $i_3 = I_{SC} = 6.034 \text{mA}$

$$\begin{aligned} R_{th} &= \frac{V_{t\Box}}{I_{sc}} \\ &= \frac{7.090}{6.034 \times 10^{-3}} \\ &= 1175 \, \Omega \\ &= 1.175 \, k\Omega \end{aligned}$$

And, $E_{th} = V_{th} = V_{OC} = 7.090 \text{V}$

For figure 4:

$$\begin{aligned} \text{Here, } I_L &= \frac{7.090}{1.175k + 1k} \\ &= 3.260 \text{mA} \end{aligned}$$

$$\begin{aligned}V_L &= I_L \times R = 3.260\text{mA} \times 1\text{k}\Omega \\ &= 3.260\text{V}\end{aligned}$$

Thus, it verifies Thevenin' theorem.

2. As the experiment was done in PSpice, the values of E ,V & R are same. Thus , there is no discrepancy.

Conclusion: In this experiment we verified Thevenin's equivalent theorem. Now we can convert two terminal linear circuits into a voltage source in series with a resistance as well as Thevenin's equivalent circuit.